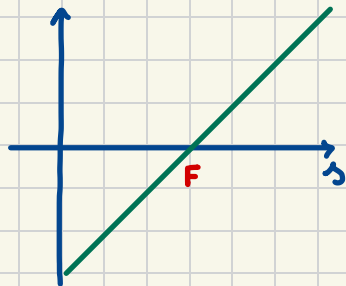
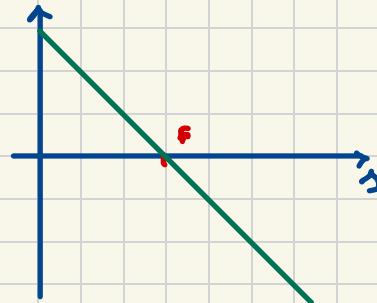


Review: An Array of Payoff Curves.

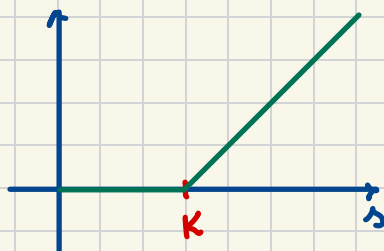
Long forward: $S(T) - F$



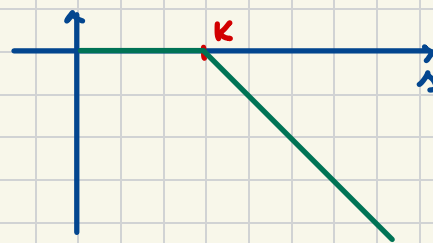
Short Forward: $F - S(T)$



Long Call: $(S(T) - K)^+$

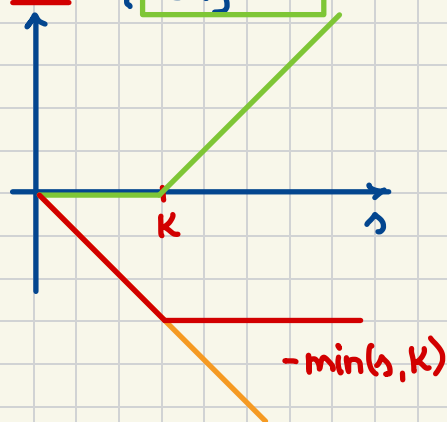


Short Call: $-(S(T) - K)^+$



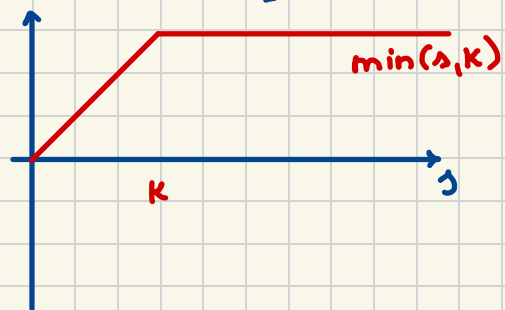
CAP.

{ Short Stock
Long Call

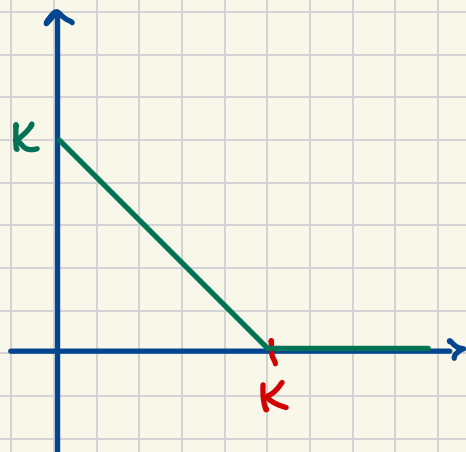


COVERED
CALL

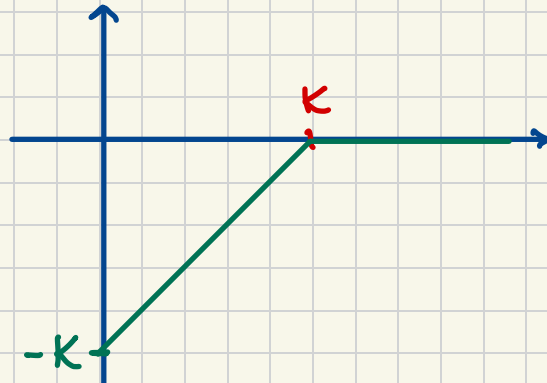
{ Short Call.
Long Stock



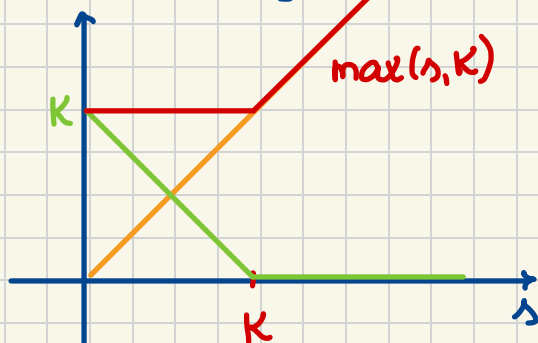
Long Put: $(K - S(T))_+$



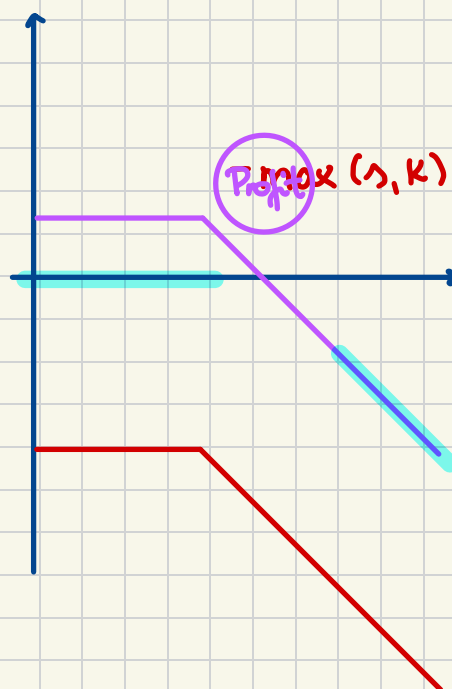
Short Put: $-(K - S(T))_+$



FLOOR { Long Stock
Long Put



COVERED PUT. { Short Put
Short Sale



Finite Probability Space.

... serve as environments for the possible paths that the stock price can take.

(e.g.)

$$S(T) \sim \begin{cases} 120 & \text{w/ prob. } \frac{1}{6} \\ 80 & \text{w/ prob. } \frac{1}{2} \\ 50 & \text{w/ prob. } \frac{1}{3} \end{cases}$$



Q: What is the expected payoff of a 105-strike put?

→:

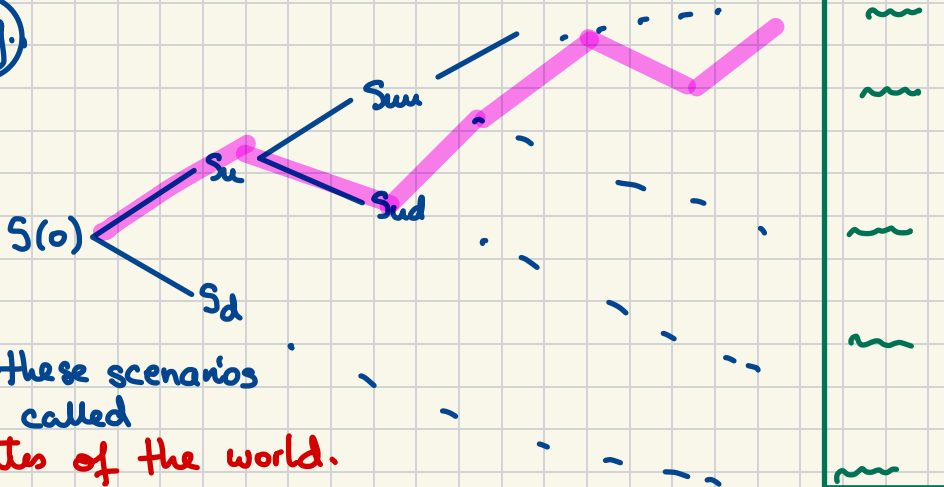
$$\mathbb{E}[(K - S(T))_+]$$

$$V_p(T) = (K - S(T))_+ \sim \begin{cases} 0 & \text{w/ prob. } \frac{1}{6} \\ 25 & \text{w/ prob. } \frac{1}{2} \\ 55 & \text{w/ prob. } \frac{1}{3} \end{cases}$$

$$\mathbb{E}[V_p(T)] = 0 \cdot \frac{1}{6} + 25 \cdot \frac{1}{2} + 55 \cdot \frac{1}{3} = \dots$$

Caveat: $\mathbb{E}[g(x)] \neq g(\mathbb{E}[x])$

(e.g.)



All these scenarios are called **states of the world.**

With finitely many states of the world:

- each one can happen, i.e., its probab > 0 and
- they exhaust all possibilities, i.e., $\sum \text{probab} = 1$.

Arbitrage Portfolio.

Def'n. An **arbitrage portfolio** is a portfolio whose PROFIT is:

- • nonnegative in ALL states of the world, w/ probab ≥ 1 , and
- • strictly positive in AT LEAST ONE state of the world w/ probab > 0 .

Unless it's specified otherwise in a specific problem or example, we assume NO ARBITRAGE.

Law of the Unique Price.

Assume that the payoffs of two static portfolios A & B are equal, i.e.,

$$V_A(T) = V_B(T)$$

random variable

T... time horizon
(fixed temporarily)

Claim.

$$V_A(0) = V_B(0)$$

Proof.

Diagnosis. ✓

Proposal. ✓

Verify. ✓ ...